-- 1. What is the total number of customers in the dataset?

```
select count(*) as total_employee from customer_churndata;
```

-- 2. How many customers have churned vs. not churned?

```
select count(*) as Chured from customer_churndata where Exited = 1;
select count(*) as Not_Chured from customer_churndata where Exited = 0;
```

-- 3. What is the average credit score of all customers?

```
select avg(CreditScore) as avg_creditScore from customer_churndata;
```

-- 4. List the number of customers from each country (Geography).

```
select Geography, count(*) as Total_customer from customer_churndata group by Geography;
```

-- 5. Find the number of male vs. female customers.

```
select Gender, count(*) as Count from customer_churndata group by Gender;
```

-- 6. How many customers have zero account balance?

```
select Balance, count(*) as Num_of_Cus from customer_churndata where Balance = 0;
```

-- 7. What is the average balance of customers who have churned?

```
select Exited, avg(Balance) as avg_bal from customer_churndata where Exited = 1;
```

-- 8. Which country has the highest churn rate?

```
select Geography, count(*) as Higest_Churn_Rate
from customer_churndata where Exited = 1
group by Geography order by Geography;
```

-- 9. What is the average tenure of customers who have not churned?

```
select Exited, avg(Tenure) as avg_Ten from customer_churndata where Exited = 0;
```

-- 10. Find the top 5 customers with the highest estimated salary who churned.

```
select CustomerId, EstimatedSalary from customer_churndata
where Exited = 1 order by EstimatedSalary desc limit 5;
```

-- 11. How many active members have more than 2 products?

```
select NumOfProducts, count(*) as ActiveCustomer
from customer_churndata
where IsActiveMember = 1 and NumOfProducts > 2
group by NumOfProducts;
```

-- 12. Compare the average balance between active and inactive customers.

```
select Exited, avg(balance) as avg_bal from customer_churndata group by Exited;
```

-- 13. What is the average estimated salary per gender?

```
select Gender, avg(EstimatedSalary) as est_salary from customer_churndata group by Gender;
```

-- 14. List the count of churned customers per tenure year.

```
select Tenure, count(*) as Chured_cus from customer_churndata
where Exited = 1 group by Tenure order by Tenure;
```

-- 15. Identify the churn rate for each product count.

```
select NumOfProducts, (sum(Exited = 1)*100/count(CustomerId)) as Churn_rate
from customer_churndata group by NumOfProducts order by NumOfProducts;
```

-- 16. What is the churn rate segmented by age groups (e.g., 18–30, 31–45, etc.)?

```
select  Case
          When Age Between 18 and 30 then "18-30"
          when Age between 31 and 45 then "31-45"
          when Age between 46 and 60 then "46-60"
          else "60+"
   end as AgeGroup,
sum(Exited = 1)*100/count(CustomerId) as ChurnRate
from customer_churndata group by AgeGroup order by AgeGroup;
```

-- 17. Which combinations of Geography and Gender have the highest churn rate?

```
select Geography, Gender, sum(Exited = 1) * 100 / count(CustomerId) as ChurnRate
from customer_churndata group by Geography, Gender order by ChurnRate desc;
```

-- 18. Find the average credit score and balance for churned customers per country.

```
select Geography, avg(CreditScore) as avg_Credit_Score, avg(Balance) as  avg_Bal
from customer_churndata
where Exited = 1 group by Geography;
```

-- 19. List the 10 most at-risk customer profiles based on high balance and churned.

```
select CustomerId, Balance from customer_churndata where Exited = 1
order by Balance Desc limit 10;
```

-- 20. What percentage of customers with no credit card have churned?

```
select sum(HasCrCard=0 and Exited=1)*100/sum(HasCrCard = 0) as NoCrCard
from customer_churndata;
```

-- 21. Which customers have a balance higher than the average balance of all churned customers?

```
select CustomerId, Balance from customer_churndata where Balance > (select avg(Balance)
from customer_churndata where Exited = 1);
```

-- 22. Which countries have a churn rate higher than the overall churn rate?

```
select Geography, sum(Exited = 1)*100/count(*) as CountryChurnRate from
customer_churndata
group by Geography having
sum(Exited = 1) * 1.0 / count(*) > (select sum(Exited = 1) * 1.0 / Count(*)
from customer_churndata);
```